

# Summary Chapter 1

|                      | Solid                       | Liquid                                  | Gas                                     |
|----------------------|-----------------------------|-----------------------------------------|-----------------------------------------|
| Shape                | Fixed                       | Not Fixed, takes shape of its container | Not Fixed, takes shape of its container |
| Volume               | Fixed                       | Fixed                                   | Not Fixed                               |
| Arrangement          | regular                     | random                                  | Random                                  |
| Separation           | Tightly packed              | Loosely packed                          | Far                                     |
| Forces of attraction | Strong Forces of attraction | Less Forces of attraction               | Very little                             |
| Motion               | Vibrate in Fixed positions  | Slide over each other                   | Randomly moving                         |
| Kinetic energy       | Very low                    | High                                    | Very high                               |
| Compressibility      | Not compressible            | Not compressible                        | Compressible                            |

**3 ways to ↑ pressure**

- ↓ volume: When volume decrease particles get closer together more frequent collisions with the inside of the wall thus, pressure increase
- ↑ temperature: particles gain Kinetic energy they move faster thus, collide more frequently
- add more gas

## 4 rules of Kinetic theory

1. Particles move all the time
2. Higher temperature → Faster movement
3. Heavier and bigger particles move slower
4. A matter's state can change by heating or cooling, but particle size doesn't change

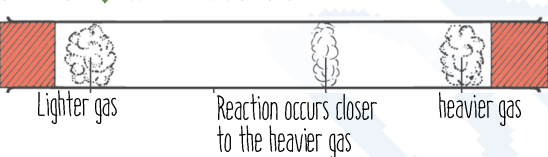
**Definition** Diffusion: Particles move randomly from areas of high concentration to areas of low concentrations

Liquid in liquid e.g. ink in water  
Gas in gas e.g. perfume in air

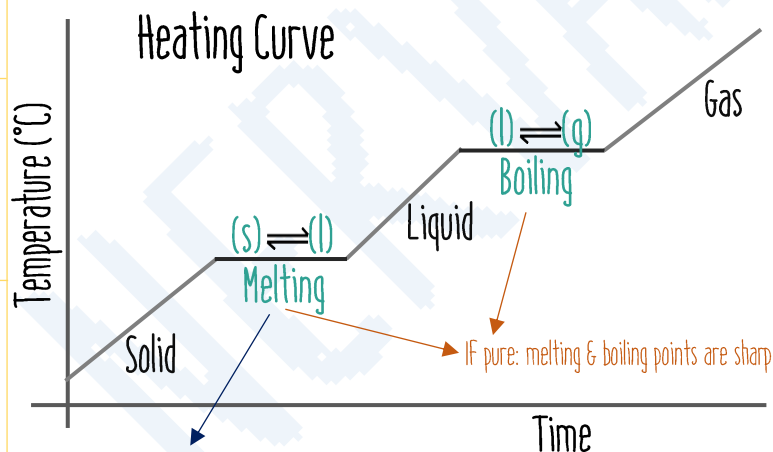
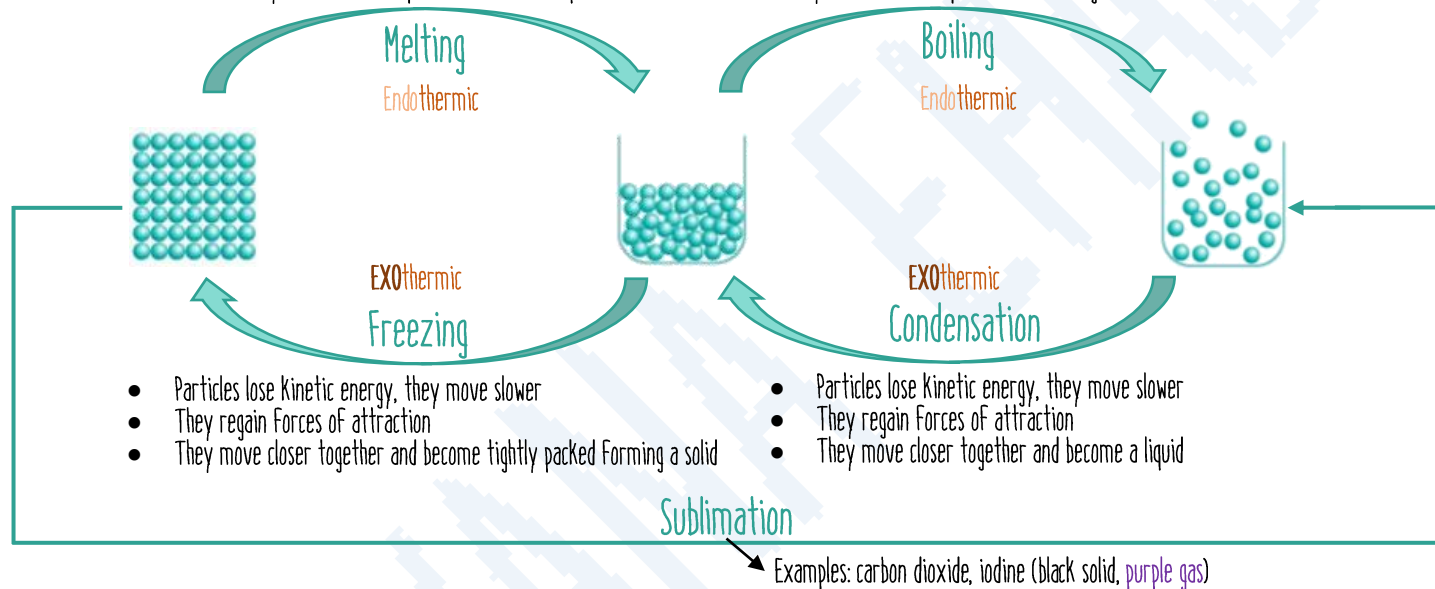
IF solid in liquid, solid must dissolve first  
IF liquid in gas, liquid must evaporate first

## 2 Factors affect rate of diffusion

1. Temperature: ↑ temperature → Fast diffusion
2. Mass: ↓ mass → Fast diffusion



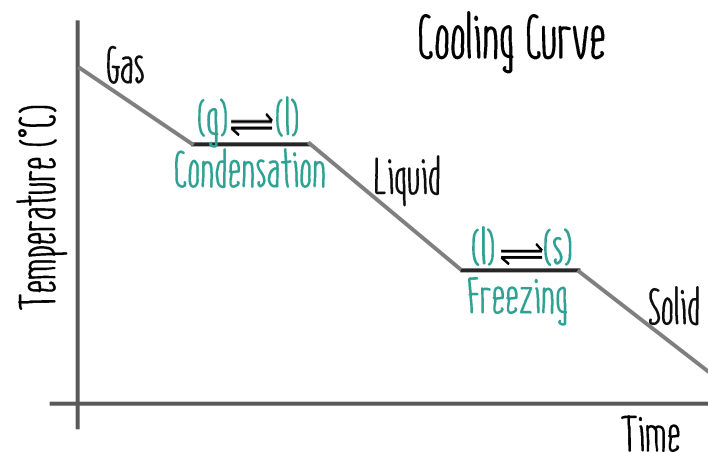
- Particles gain Kinetic energy, they vibrate faster
- Until energy is enough to overcome forces of attraction
- They move further apart and become a liquid
- Particles gain Kinetic energy, they move faster
- Until energy is enough to overcome forces of attraction
- They move further apart to become a gas, then bubble rise to the surface



Temperature doesn't change for some time during melting & boiling because energy is being used to overcome attractive forces between particles

IF impurities are present:

- melting and boiling points are not sharp → range
- Melting point ↓
- Boiling point ↑
- The more impurities, the bigger the changes in m.pt & b.pt



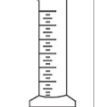
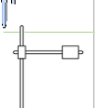
## Evaporation

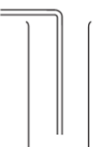
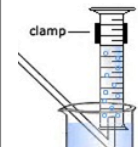
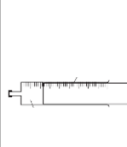
Slow  
Occurs at all temperatures (between melting and boiling point)  
Occurs only at the surface

## Boiling

Rapid  
Occurs only at boiling point  
Occurs throughout the liquid

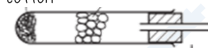
# Summary Chapter 2

| Apparatus                                                                        | Name & Use                                                                                             | Apparatus                                                                           | Name & Use                                       |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------|
|  | Stopwatch: to measure time                                                                             |    | Crucible with lid: to heat solids                |
|  | Measuring cylinder to measure volume to the nearest 1 cm³. Easy & quick to use                         |    | Evaporating dish: to evaporate solutions         |
|  | Conical flask: to put liquids in it, for mixing & swirling                                             |    | Watch glass: to hold solids, to cover a beaker   |
|  | Gas syringe: to measure volume of gas                                                                  |    | Spatula: to transfer solids                      |
|  | Pipette: to measure a fixed volume of liquid accurately. Sizes: 10, 25, 50 cm³. Slow to use            |    | Pestle to crush solids<br>Mortar                 |
|  | Burette: measures volume of liquid to the nearest 0.1 cm³. Slow to use. Used for titration experiments |    | Separating funnel to separate immiscible liquids |
|  | Retort stand: to hold other apparatus                                                                  |    | Sandpaper: to clean metal surfaces               |
|                                                                                  |                                                                                                        |    | Glass rod: to stir                               |
|                                                                                  |                                                                                                        |    | Tongs: to lift hot items                         |
|                                                                                  |                                                                                                        |    | Beaker: to store, mix, & heat liquids            |
|                                                                                  |                                                                                                        |   | Thermometer: to measure temperature              |
|                                                                                  |                                                                                                        |  | Wire gauze: to support apparatus                 |
|                                                                                  |                                                                                                        |  | Tripod stand: to support apparatus               |
|                                                                                  |                                                                                                        |  | Bunsen burner: to heat substances                |

| Downward delivery                                                                  | Upward delivery                                                                     | Over water                                                                          | Gas syringe                                                                         |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| gas is heavier than air<br>$M_r > 29$                                              | gas is lighter than air<br>$M_r < 29$                                               | To measure volume of gas that's insoluble                                           | To measure volume of any gas                                                        |
|  |  |  |  |

## Separating chemicals:

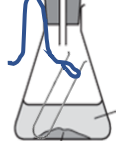
- Hold liquid in ceramic wool or cotton



- Separate a liquid using separating funnel

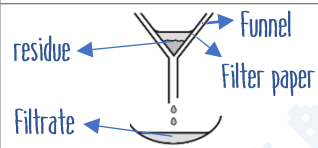
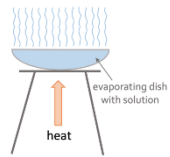
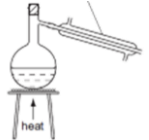
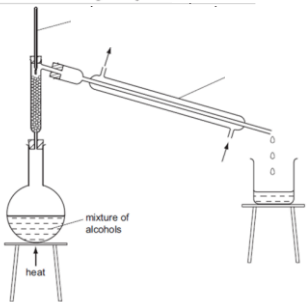
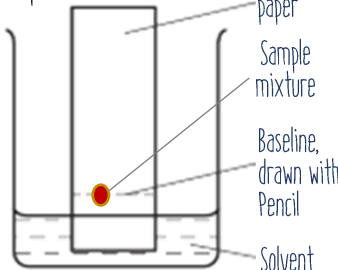


- Separate a liquid by putting it in a test tube and hanging it using a thread



- Place a divider inside a conical flask to separate 2 chemicals



|                         | Method used For                                                                                                                                              | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filtration              | Obtain an insoluble solid from a mixture<br>E.g., Sand from sea water                                                                                        |  <p>Decantation: pouring off the liquid leaving the solid behind</p> <p>Centrifugation: heavier solid settles at the bottom due to spinning the tube at high speed</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Evaporation             | Obtain a soluble solid from a solution.<br>If it's a salt, the anhydrous form will be obtained<br>E.g., salt from sea water                                  | <p>Heat the mixture in an evaporating dish. Solvent will evaporate. Solids will be left behind.</p> <p><b>Definitions</b><br/> <u>Solute</u>: the substance dissolved in a solvent, to make a solution<br/> <u>Solvent</u>: the liquid in which a solute is dissolved, to make a solution<br/> <u>Solution</u>: a mixture obtained when one or more solutes are dissolved in a solvent</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Crystallization         | Obtain a soluble solid from a solution, in the hydrated form, the less soluble solute will crystallize first<br>E.g., most soluble salts                     | <p>Heat until crystallization point, cool, filter out the crystals, dry them on a filter paper</p> <p><b>Definitions</b><br/> <u>Crystallization</u>: the process in which crystals form as a saturated solution cools down<br/> <u>Saturated solution</u>: a solution that contains the maximum concentration of a solute dissolved in the solvent at a specific temperature<br/> <u>Solubility</u>: the amount of solute that dissolves in 100 grams of a solvent, at a given temperature</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Simple distillation     | Obtain a solvent from a solution<br>E.g., obtain pure water from a solution                                                                                  | <p>Make sure the flask is sealed at the top with a bung to prevent loss of vapour</p> <p>Condenser is filled with cold water from bottom up, so water would fill it up</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Fractional distillation | Obtain a liquid from a mixture of miscible liquids, (or obtain a gas from a mixture of gases, e.g. air)<br>E.g., Ethanol from a mixture of ethanol and water | <ul style="list-style-type: none"> <li>Method works because the liquids have different boiling points</li> <li>The liquid with lower boiling point will be obtained first</li> <li>Fractionating column provides high surface area to condense liquids with higher boiling points</li> <li>Thermometer helps tell you which liquid is being collected</li> </ul> <p>Fractional distillation is used to separate:</p> <ol style="list-style-type: none"> <li>Liquefied air</li> <li>Petroleum</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Chromatography          | Separate a mixture of soluble substances<br>E.g., A mixture of colors, mixture of amino acids, mixture of simple sugars, components of ink                   | <ul style="list-style-type: none"> <li>On a chromatography paper draw a line near the bottom with pencil.</li> <li>On this line, place few drops of the sample.</li> <li>Place the chromatography paper in a beaker containing a suitable solvent so that the pencil line is above the solvent level.</li> <li>Allow the solvent to run up the paper to separate the substances.</li> <li>Individual spots will indicate the number of colored substances in the sample.</li> <li>If the sample is colorless, use a locating agent to reveal the spots</li> </ul> <p>Purpose of chromatography is to separate &amp; identify</p> <p>To identify:</p> <ol style="list-style-type: none"> <li>Measure <math>R_f</math> value</li> <li>Compare sample spot level with known substances</li> </ol> <p>Solvent level should be lower than baseline, to prevent wash away of sample</p> <p>Baseline drawn in pencil because pen would get separated</p> <p>Where the solvent reaches (near the top) is called: Solvent Front</p> <p><math>R_f = \frac{\text{distance traveled by spot}}{\text{distance traveled by solvent}}</math></p>  |

## Safety measures:

- No running, eating, or drinking, tie hair
- Wear lab coat, gloves, & goggles
- Do not look into an apparatus containing chemicals
- Point test tube away when heating
- Do **not** heat alcohols using Bunsen burner, because they are flammable. Use electric heater or water bath
- Use **Fume cupboards** or well-ventilated area when dealing with toxic gases
- Prevent suck-back by removing delivery tube from the liquid before stopping heating



Corrosive



Flammable

# Summary Chapter 3

Net charge = 0 ( $p^+ = e^-$ )

## Atom

|          | Protons        | neutrons       | electrons                 |
|----------|----------------|----------------|---------------------------|
| Charge   | +1             | no charge      | -1                        |
| Mass     | 1 unit         | 1 unit         | 1/1840                    |
| Location | inside nucleus | inside nucleus | outside nucleus in shells |

**Definition** Isotopes: different atoms of the same element that have the same number of protons but different numbers of neutrons

## Electronic configuration

1<sup>st</sup> shell → 2 max  
 2<sup>nd</sup> shell → 8 max  
 3<sup>rd</sup> shell → 18 max

Outer shell → 8 max (except 1<sup>st</sup> shell)  
 8 electrons in outer shell → stable  
 except shell #1: 2 electrons → stable

## Periodic table

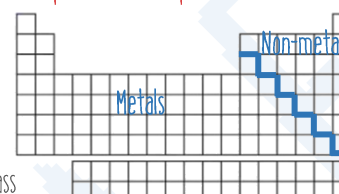
**Definition** Element: a pure substance containing only one type of atom, and cannot be split into anything simpler

**Rules** Elements are arranged in order of proton #

Columns - Groups → determines # of outer electrons (valency electrons), except H (not in Group I) & He (not in Group II)

Rows - periods determines # of shells

19 → Proton # = atomic # → identifies the element  
 K → Symbol → 1 or 2 letter, 1st letter always uppercase  
 potassium → Name  
 39 → Mass # = nucleon # =  $p + n$  = Ar = relative atomic mass



Average  $A_r$  of isotopes =  $(\text{mass})(\% \text{ abundance}/100) + (\text{mass})(\% \text{ abundance}/100)$

| Groups  | I  | II | III | IV   | V  | VI | VII | VIII |
|---------|----|----|-----|------|----|----|-----|------|
| Valency | +1 | +2 | +3  | +/-4 | -3 | -2 | -1  | 0    |

## Metals vs. non-metals

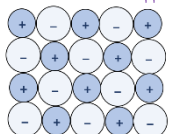
|                     | Metals                                                                                                      | non-metals                                                                              |
|---------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Physical properties | Good conductors<br>High m.pt & b.pt<br>Hard, strong, high density<br>Malleable & ductile<br>Shiny, sonorous | Non-conductors<br>Low m.pt & b.pt<br>Soft, low density<br>Brittle<br>Dull, non-sonorous |
| Chemical properties | Loses electrons,<br>Forms +ve ions (cations)                                                                | Gains electrons,<br>Forms -ve ions (anions)                                             |

**Definition** molecules: a unit of two or more atoms bonded together

**Definition** compound: containing 2 or more different atoms, chemically bonded together

## Ionic

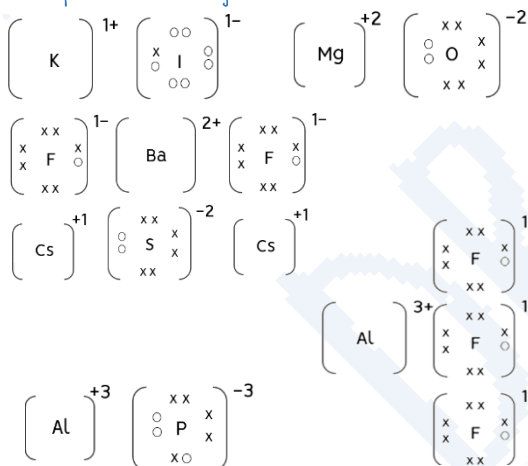
**Definition** transfer of electrons from metal to non-metal forming strong electrostatic attraction between oppositely charged



### Properties

- ↑ m.pt & b.pt  
Due to strong electrostatic attractive forces b/w oppositely charged ions → requires lot of energy to overcome
- Soluble in water
- Conduct electricity when molten or dissolved  
Due to free ions

### Examples of dot-cross diagrams



## Simple covalent

**Definition** Formed when a pair of electrons is shared between two non-metals leading to noble gas electronic configurations ions arranged regularly in a giant lattice structure

### Properties

- ↓ m.pt & b.pt  
Due to the weak intermolecular forces of attraction
- Insoluble in water  
Except  $NH_3$ , acids, alcohols
- Doesn't conduct electricity  
No free electrons, no free ions

| Group IV                                     | Group VI                                    | Group V                                     | Group VII                                                                          |
|----------------------------------------------|---------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------|
| 4 outer electrons<br>4 shared<br>0 unshared  | 6 outer electrons<br>2 shared<br>4 unshared | 5 outer electrons<br>3 shared<br>2 unshared | 7 outer electrons<br>1 shared<br>6 unshared                                        |
| Bonding options                              | Bonding options                             | Bonding options                             | Bonding options                                                                    |
| $\begin{array}{c}   \\ -C- \\   \end{array}$ | $O=$                                        | $N \equiv$                                  | $Cl-$                                                                              |
| $\begin{array}{c}   \\ -C= \\   \end{array}$ | $-O-$                                       | $-N=$                                       | Hydrogen<br>1 outer electrons<br>1 shared<br>0 unshared<br>Bonding options<br>$H-$ |
| $=C=$                                        |                                             | $-N-$                                       |                                                                                    |
| $-C \equiv$                                  |                                             |                                             |                                                                                    |

## Macromolecules

### Properties

- ↑ m.pt & b.pt  
Due to the large # of covalent bonds, requires a lot of energy to break
- Insoluble in water.  
Due to the large structure
- Not conduct electricity.  
No ions, no free electrons  
Except graphite: each carbon forms 3 covalent bonds, the 4<sup>th</sup> electron is free
- Hard.  
Due to the giant tetrahedral structure  
Except graphite: soft due to weak forces of attraction between layers → can slide over each other

**Describe** each carbon atom is covalently bonded to 4 other carbon atoms tetrahedrally

**Uses** cutting & drilling, cz hard

**Describe** each carbon atom is covalently bonded to 3 other carbon atoms forming layers

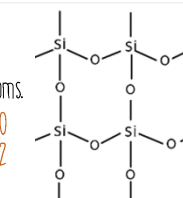
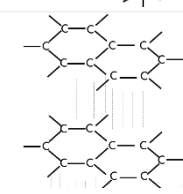
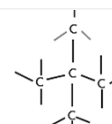
**Uses**

- lubricant & pencil, cz soft
- Electrodes, cz conducts electricity

**Describe** each silicon atom is covalently bonded to 4 O atoms tetrahedrally, and each O atom is bonded to 2 Si atoms. Structure is similar to that of diamond

**Use** make glass, cz hard

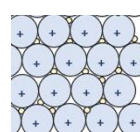
Ratio Si : O  
1 : 2



## Metallic

### Describe

Lattice of regularly arranged cations surrounded by a sea of delocalized electrons. There's a strong attractive forces b/w cations and electrons



### Properties of metals

- Conduct electricity  
due to the free delocalized electrons
- Malleable  
Layers of regularly arranged cations can slide over each other when force is applied
- ↑ m.pt & b.pt  
Due to the strong attractive forces b/w cations and electrons

**Definition** Alloy: mixture of metal & another element

Brass: Cu + Zn  
Bronze: Cu + Sn  
Steel: Fe + C

### Properties of alloys

- Harder  
due to the different size atoms which makes sliding of layers harder
- Resists corrosion
- Better appearance

# Summary Chapter 4

Relative molecular mass -  $M_r$ : The sum of relative atomic masses.  
Calculated by adding up the  $A_r$  of all atoms in the compound

Calculation

$$\% \text{ of an element in a compound} = \frac{A_r \text{ of the element} \times \text{number of atoms} \times 100}{M_r \text{ of the compound}}$$

| Name      | Formula            | Name      | Formula            | Name     | Formula            |
|-----------|--------------------|-----------|--------------------|----------|--------------------|
| sulfate   | $\text{SO}_4^{2-}$ | sulfite   | $\text{SO}_3^{2-}$ | Silver   | $\text{Ag}^+$      |
| nitrate   | $\text{NO}_3^{-1}$ | nitrite   | $\text{NO}_2^{-1}$ | Ammonium | $\text{NH}_4^{+1}$ |
| phosphate | $\text{PO}_4^{3-}$ | phosphite | $\text{PO}_3^{3-}$ | Zinc     | $\text{Zn}^{2-}$   |
| carbonate | $\text{CO}_3^{2-}$ |           |                    | Lead     | $\text{Pb}^{+2}$   |
| hydroxide | $\text{OH}^{-1}$   |           |                    | Nickel   | $\text{Ni}^{+2}$   |

Writing Formula



Definition A mole of any substance is its  $M_r$  (or  $A_r$ ) expressed in grams

Rules

Balancing #s = mole ratio

Knowing moles of 1 substance, you can find the moles of all other substances in a balanced equation using mole ratio

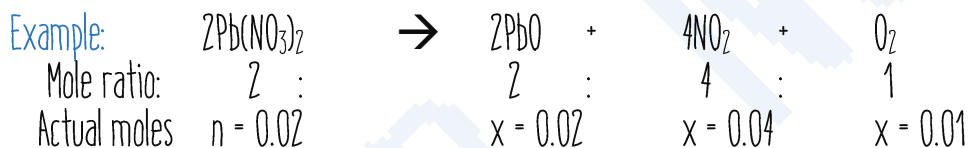
Finding oxidation state

Example: Find oxidation state of N in  $\text{NO}_3^{-1}$



There are 3 atoms of oxygen, so multiply by 3

$$\begin{aligned} x & \\ +x - 6 &= -1 & -(2)(3) &= -1 & x &= +5 \end{aligned}$$



Calculation

$$n = \frac{\text{mass in grams}}{M_r} \quad \text{Unit: g/mol}$$

$M_r \rightarrow \text{Mass of 1 mol}$

If both reactants are given, Find which is excess & which is limiting

- 3 steps
- Find moles of each reactant
  - Divide by mole ratio
  - Compare: bigger is excess

Limiting reactant determines the amount of products

Classic question: given mass of 1 substance, Find mass of another substance in a balanced equation



Mass =  $\checkmark$  mass = ?

3 steps

- Find moles of the given substance
- Find moles of the unknown substance
- Find mass of the unknown substance

Rule

Mass of reactants is equal to mass of products

1 mole =  $6.02 \times 10^{23}$  particles (Avogadro's constant)

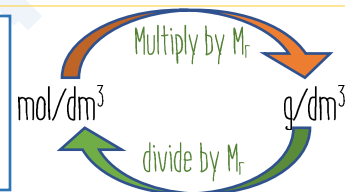
Calculation

$$n = \frac{C \times V}{1000}$$

For aqueous solutions ONLY

1 dm<sup>3</sup> = 1000 cm<sup>3</sup>

Converting concentration units



Classic question: given C & V of 1 substance, Find the C of another substance in a balanced equation



C =  $\checkmark$  C = ?  
V =  $\checkmark$  V =  $\checkmark$

3 steps

- Find moles of the given substance
- Find moles of the unknown substance
- Find C of the unknown substance

Calculation

$$V = \frac{24n}{1000}$$

For gases ONLY

1 mole of any gas occupies 24 dm<sup>3</sup> at r.t.p

Rule

Gas volume ratio = mole ratio = balancing numbers

Definition

Empirical Formula: the simplest whole number mole ratio of the different elements in a compound

Classic questions: given masses or percentages of the different elements, Find empirical Formula

2 steps

- Find moles of the different elements (mass/ $A_r$ )
- Simplify by dividing by the smallest ratio

Given empirical Formula &  $M_r$ , Find the molecular Formula

Calculations

$$x = \frac{M_r}{\text{empirical mass}}$$

integer

$$\text{MF} = x [\text{EF}]$$

Molecular Formula integer Empirical Formula

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{Theoretical yield}} \times 100$$

$$\% \text{ purity} = \frac{\text{mass of pure}}{\text{Total mass}} \times 100$$



# Summary Chapter 5

**Definition** Electrolysis: the decomposition (or break down) of a substance by the passage of electricity

Electrolysis is an endothermic process because it absorbs energy

Electricity is conducted because of the:

- Free electrons in wires and electrodes
- Free ions in the electrolyte

In electrolysis, electrical energy is converted to chemical energy

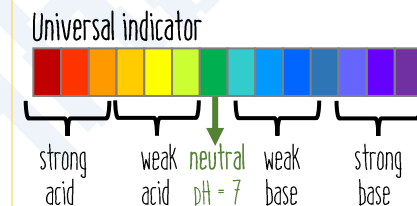
## Electrolysis of molten

| At cathode (-ve electrode)                                                                                                                                             | At anode (+ve electrode)                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Metal ions + e <sup>-</sup> → Metal element<br>e.g. $Pb^{2+}_{(l)} + 2e^{-} \rightarrow Pb_{(l)}$                                                                      | Nonmetal ion - e <sup>-</sup> → Non-metal element<br>e.g. $2Br^{-}_{(l)} - 2e^{-} \rightarrow Br_{2(g)}$                                                                                                                                                                                                                                    |
| Observation<br>Deposits of metal<br>All metals are silvery grey, except<br><ul style="list-style-type: none"> <li>Gold → yellow</li> <li>Copper → red brown</li> </ul> | Observation: Bubbles of gas<br><ul style="list-style-type: none"> <li>Fluorine → pale-yellow gas</li> <li>Chlorine → yellow-green gas</li> <li>Bromine → red-brown liquid</li> <li>Iodine → black solid, purple gas</li> <li>Oxygen, nitrogen, hydrogen → Colorless gas</li> <li>Carbon dioxide, carbon monoxide → Colorless gas</li> </ul> |
| Gain of electrons → reduction                                                                                                                                          | Loss of electrons → oxidation                                                                                                                                                                                                                                                                                                               |

## Solubility rules:

- All Group I & ammonium salts  $NH_4^+$  are soluble
- All nitrates  $NO_3^-$  are soluble
- All halides are soluble except  $Ag^+$  &  $Pb^{2+}$
- All sulfates  $SO_4^{2-}$  are soluble except  $Ca^{2+}$ ,  $Ba^{2+}$  &  $Pb^{2+}$
- All carbonates are insoluble except Group I &  $NH_4^+$
- All hydroxides are insoluble except Group I &  $NH_4^+$ , Calcium hydroxide is sparingly soluble
- All oxides are insoluble except Group I,  $NH_4^+$ ,  $Ca^{2+}$ ,  $Mg^{2+}$

| Indicator       | acid      | neutral   | base   |
|-----------------|-----------|-----------|--------|
| Litmus          | red       | no change | blue   |
| Methyl orange   | red       | orange    | yellow |
| Thymolphthalein | colorless | colorless | blue   |



| Inert electrodes                                      | active electrodes                                                                                     |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Graphite or platinum                                  | not graphite or platinum                                                                              |
| Cathode: less reactive cation discharged              | cathode: less reactive cation discharged                                                              |
| Anode: $OH^-$ discharged except conc. halide          | anode: metal dissolves producing cations                                                              |
| Electrolyte changes (e.g. $Cu^{2+}$ blue color Fades) | electrolyte stays the same: cations deposited at cathode gets replaced by cations produced from anode |

More reactive  
→ more likely to be ions

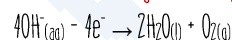
## Reactivity series

K  
Na  
Ca  
Mg  
Al  
Zn  
Fe  
Pb  
H  
Cu  
Hg  
Ag  
Au  
Pt

Less reactive  
→ less likely to be ions

## Electrolysis of aqueous solutions

- Rules**
- Less reactive cation gets discharged at cathode
  - $OH^-$  gets discharged at anode except concentrated halide



Electrolysis of conc. NaCl produces 3 products:

| $Cl_2$ (at anode)              | $H_2$ (at cathode)                     | NaOH (remains)              |
|--------------------------------|----------------------------------------|-----------------------------|
| bleach<br>plastic disinfectant | Fuel<br>margarine<br>make HCl & $NH_3$ | soap<br>paper<br>detergents |

Test For  $Cl_2$ : bleaches damp litmus paper  
Test For  $H_2$ : pops with a lighted splint  
Test For  $O_2$ : relights a glowing splint

## Electrolysis using active electrodes

### Purification

- Rules**
- Anode: impure metal:  $M - Ze^{-} \rightarrow M^{Z+}$
  - Cathode: pure:  $M^{Z+} + Ze^{-} \rightarrow M$
  - Electrolyte: soluble salt of the metal being purified

### electroplating

- Rules**
- Anode: pure metal
  - Cathode: Object
  - Electrolyte: soluble salt of the metal

Purpose: 1. Give shiny appearance, 2. Prevent corrosion

Considerations:

- Clean object with sandpaper
- Completely immerse object
- Rotate object for even coating

## Extraction of aluminium

**Ore:** Bauxite,  $Al_2O_3$   
**Cryolite,**  $Na_3AlF_6$  is used to:

- Reduce melting point, thus save energy & cost
- Increase conductivity
- Act as solvent

| At graphite cathode (-ve electrode)                                                      | At graphite anode (+ve electrode)                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Al^{3+}_{(l)} + 3e^{-} \rightarrow Al_{(l)}$                                            | $2O^{2-}_{(l)} - 4e^{-} \rightarrow O_{2(g)}$                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Observation: Silvery grey aluminium sinks to the bottom<br>Gain of electrons → reduction | Observation: Bubbles of colorless gas<br>Loss of electrons → oxidation<br>Four Gases are produced at anode:<br><ol style="list-style-type: none"> <li>Oxygen, <math>O_2</math>, due to oxidation of <math>O^{2-}</math></li> <li>Carbon dioxide, <math>CO_2</math>, due to burning of anode</li> <li>Carbon monoxide, CO, due to incomplete combustion of anode</li> <li>Fluorine, <math>F_2</math>, due to oxidation of <math>F^-</math> from cryolite</li> </ol> |

# Summary Chapter 6

## Exothermic

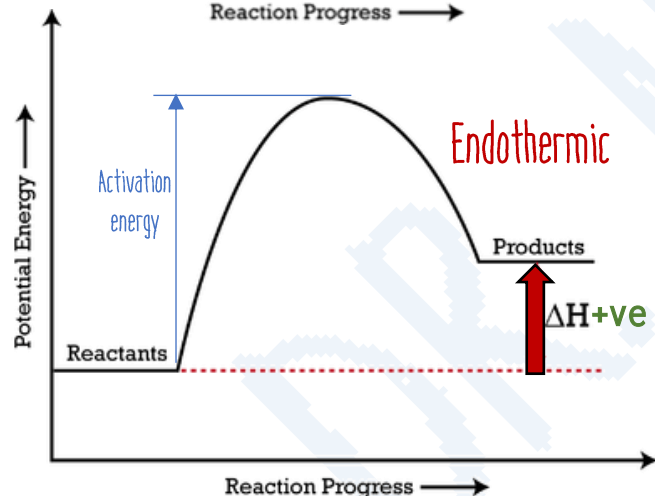
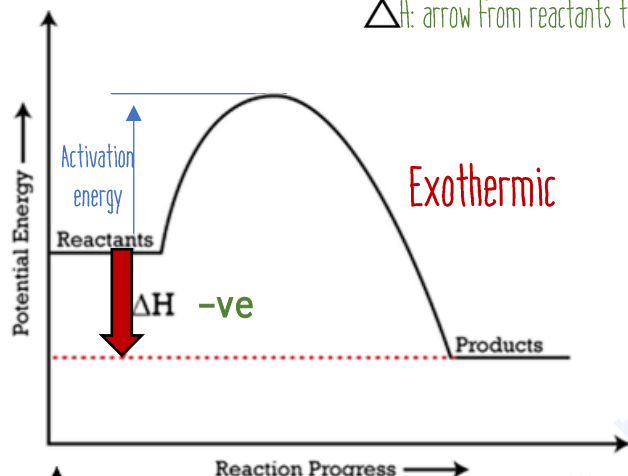
Energy is released, heat given out  
Reaction mixture becomes hot  
Energy released when bonds Formed is more than energy absorbed to break bonds  
Products have lower energy level than reactants  
Examples:  
Neutralization, Displacement, Combustion, Fermentation, Hydrogen-oxygen Fuel cell  
Condensation, Freezing

## Endothermic

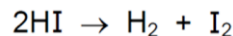
Energy is absorbed, heat taken in  
Reaction mixture becomes cold  
Energy absorbed to break bonds is greater than energy released to form bonds  
Products have higher energy level than reactants  
Examples:  
thermal decomposition, electrolysis, cracking, photosynthesis  
melting, boiling

## Energy level diagrams

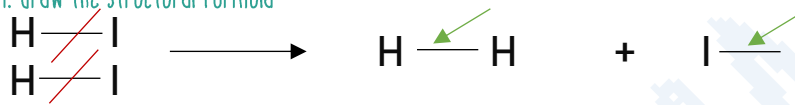
E<sub>a</sub>: arrow From reactants to peak  
 $\Delta H$ : arrow From reactants to products



## Exam question style #1: Classic, asks For $\Delta H$ (enthalpy change)



Step #1: draw the structural Formula

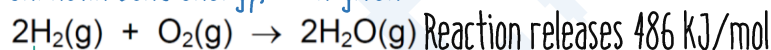


Step #2: in a table, calculate bonds broken & bonds made

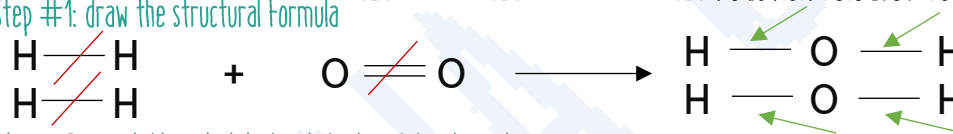
| Bonds broken                      | Bonds made                                                                |
|-----------------------------------|---------------------------------------------------------------------------|
| $\text{H-I} = 300 \times 2 = 600$ | $\text{H-H} = 440 \times 1 + \text{I-I} = 150 \times 1 = 440 + 150 = 590$ |

Step #3: bond breaking - bond making  $\Delta H = 600 - 590 = +10$

## Exam question style #2: unknown bond energy, $\Delta H$ given



Step #1: draw the structural Formula

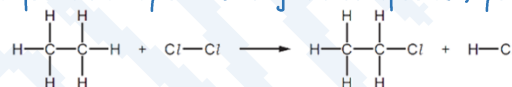


Step #2: in a table, calculate bonds broken & bonds made

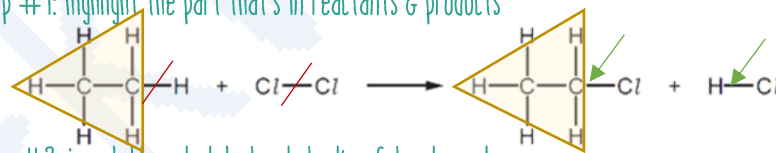
| Bonds broken                                                                 | Bonds made                         |
|------------------------------------------------------------------------------|------------------------------------|
| $\text{H-H} = 436 \times 2 = 872 + \text{O-O} = x, \text{ broken} = 872 + x$ | $\text{H-O} = 464 \times 4 = 1856$ |

Step #3: bond breaking - bond making  $\Delta H = 872 + x - 1856 = -486$  thus  $x = 498$

## Exam question style #3: organic compounds, you don't break all bonds & make them again



Step #1: highlight the part that's in reactants & products



Step #2: in a table, calculate bonds broken & bonds made

| Bonds broken                                  | Bonds made                                    |
|-----------------------------------------------|-----------------------------------------------|
| $\text{C-H} = 410 + \text{Cl-Cl} = 240 = 650$ | $\text{C-Cl} = 340 + \text{H-Cl} = 430 = 770$ |

Step #3: bond breaking - bond making  $\Delta H = 650 - 770 = -120$

## Calculations

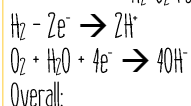
| bond | bond energy kJ/mol |
|------|--------------------|
| H-H  | 440                |
| I-I  | 150                |
| H-I  | 300                |

| bond | bond energy kJ/mol |
|------|--------------------|
| H-O  | 464                |
| H-H  | 436                |
| O-O  | ?                  |

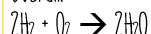
| bond  | bond energy kJ/mol |
|-------|--------------------|
| C-Cl  | 340                |
| C-H   | 410                |
| Cl-Cl | 240                |
| H-Cl  | 430                |

## hydrogen

$\text{H}_2\text{-O}_2$  Fuel cell



Overall:

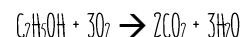


Advantages: no pollution, produce only  $\text{H}_2\text{O}$ ,  $\text{H}_2$  is easy to make

Disadvantages:  $\text{H}_2$  is flammable  $\rightarrow$  difficult to store

## Fuels Any substance that provides energy

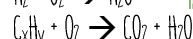
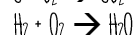
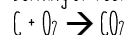
### ethanol



Used in car engines

### Combustion reactions

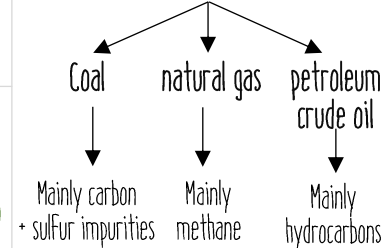
Burning of Fuel in oxygen, it's exothermic



Complete: excess  $\text{O}_2$ , produces  $\text{CO}_2$

Incomplete: insufficient  $\text{O}_2$ , produces  $\text{CO}$

## Fossil Fuels



## Physical change

## Chemical change

No new substance is Formed  
Easy to reverse  
Changes in state  
Dissolving in a solvent

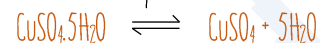
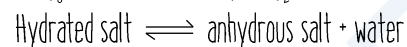
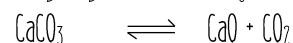
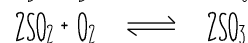
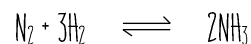
new substance is Formed  
not easily reversed  
neutralization, redox, combustion  
displacement, decomposition

# Summary Chapter 7

## Examples of catalysts:

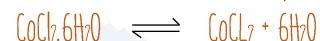
1. **MnO<sub>2</sub>**:  $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
2. **Iron**:  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$
3. **V<sub>2</sub>O<sub>5</sub>**:  $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$
4. **Platinum**: catalytic converters
5. **Nickel**:  $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$
6. **Nickel**: addition of hydrogen to alkenes
7. **Conc. H<sub>2</sub>SO<sub>4</sub>**: esterification
8. **Al<sub>2</sub>O<sub>3</sub>**: cracking
9. **H<sub>3</sub>PO<sub>4</sub>**: addition of steam to alkene
10. **Yeast (enzyme)**: Fermentation
11. **Chlorophyll**: photosynthesis

## Examples of reversible reactions:



Observation of Forward rx: blue to white, condensed water at top

Observation of backward rx: white to blue, tube gets warm



**Definition** Equilibrium: a reversible reaction in which the rate of Forward reaction is equal to the rate of backward reaction and amounts of all substances remain constant

## Effect of concentration on equilibrium:

- if a substance is added, equilibrium shifts to the direction that gets rid of this substance.
- If a substance is removed, equilibrium shifts to the side that forms that substance

## Effect of temperature on equilibrium:

- If exothermic: ↑ temperature shifts equilibrium backward, ↓ temperature shifts equilibrium Forward
- If endothermic: ↑ temperature shifts equilibrium Forward, ↓ temperature shifts equilibrium backward

## Effect of pressure on equilibrium:

- ↑ pressure shifts equilibrium to the side with less gas moles
- ↓ pressure shifts equilibrium to the side with more gas moles

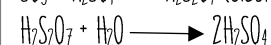
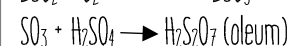
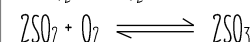
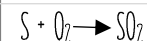
## Haber process

To make ammonia  
 $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$

450C, 200 atm, Fe (catalyst)

## Contact process

to make sulfuric acid



450C, 2 atm, V<sub>2</sub>O<sub>5</sub> (catalyst)

Why not use higher temperature:

equilibrium will shift backward

Why not use lower temperature:

reaction will be slow

Why not use higher pressure:

expensive & dangerous

Why not use lower pressure:

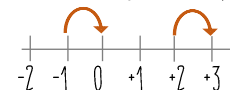
equilibrium will shift backward

## Oxidation

Gain of oxygen

Loss of electrons

Increase in oxidation number



## Reduction

loss of oxygen

gain of electrons

decrease in oxidation number



## Oxidizing agents

**KMnO<sub>4</sub>** purple → colorless

**K<sub>2</sub>Cr<sub>2</sub>O<sub>7</sub>** orange → green

Oxygen, H<sub>2</sub>O<sub>2</sub>, HNO<sub>3</sub>

less reactive cations

reactive non-metals

## Reducing agents

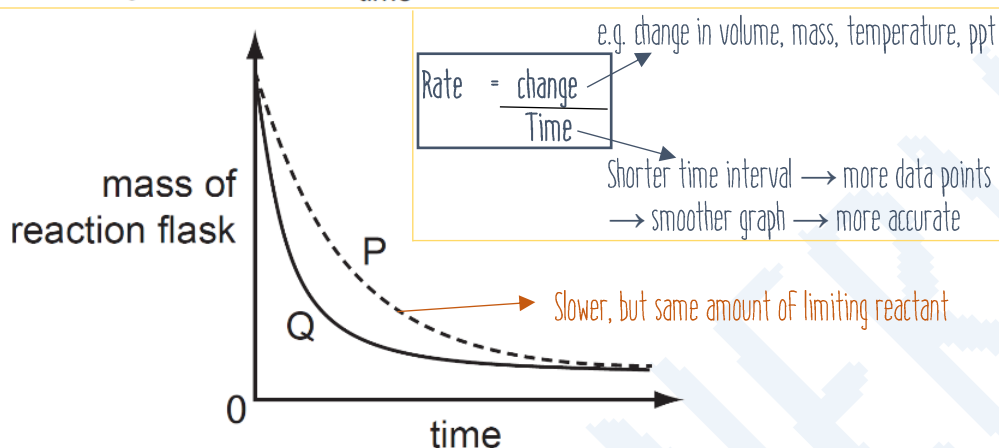
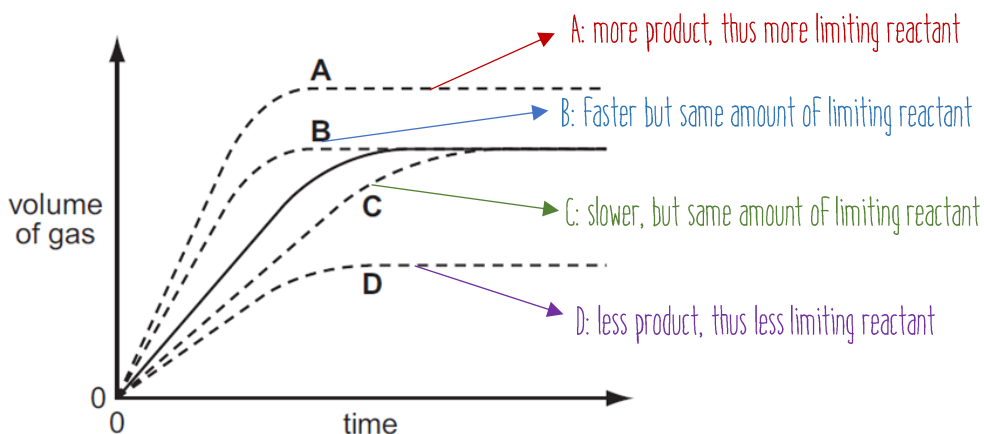
KI colorless → brown

Carbon, carbon monoxide

Hydrogen, H<sub>2</sub>S

Reactive metals

Non-reactive anions



2 conditions are needed for a reaction to occur: 1. Collision, 2. Collision with sufficient energy

## Factors affecting rate of reaction:

1. **Concentration**: more particles per unit volume, thus more frequent collisions per unit time
2. **Temperature**: particles have more kinetic energy, they move faster, so they collide more frequently per unit time. Particles also have energy more than activation energy thus more successful collisions
3. **Pressure**: pressure ↑ volume ↓ so particles get closer together, they collide more frequently per unit time
4. **Surface area**: more particles exposed, so more frequent collisions per unit time
5. **Stirring**: more frequent collisions
6. **Catalyst**: increase rate without being used up. Catalyst decreases the activation energy of the reaction
7. **Light**: e.g. photosynthesis, substitution of alkanes with halogen
8. **Reactivity**: more reactive metals → faster reaction

# Summary Chapter 8

Preparation of salts

Salt

Salt soluble

Salt insoluble

Acid + base reaction

Start with 2 aqueous solutions

Base insoluble

base soluble

Neutralization method

Titration method

Precipitation Method

1. Add excess base to acid
2. Heat (if insoluble MO or MOH)
3. Filter to remove excess base
4. Heat filtrate to crystallization point
5. Cool, filter off crystals, dry

1. Place 25 cm<sup>3</sup> of acid in conical flask
2. Add methyl orange → red
3. Using burette, titrate with alkali
4. until red turns orange
5. Record volume of alkali used
6. Repeat without indicator using same volumes
7. Heat to crystallization point, cool, filter, dry

1. Mix the 2 aqueous solutions in a beaker
2. Ppt will form
3. Filter
4. Wash residue with distilled water
5. Dry between filter paper

| Cation                       | + NaOH(aq)                                         | + NH <sub>3</sub> (aq)                              |
|------------------------------|----------------------------------------------------|-----------------------------------------------------|
| NH <sub>4</sub> <sup>+</sup> | ammonia produced on warming                        | -                                                   |
| Ca <sup>2+</sup>             | white ppt., insoluble in excess                    | no ppt. or very slight white ppt.                   |
| Cu <sup>2+</sup>             | light blue ppt., insoluble in excess               | light blue ppt., soluble in excess → dark blue sol. |
| Fe <sup>2+</sup>             | green ppt., insoluble in excess                    | green ppt., insoluble in excess                     |
| Fe <sup>3+</sup>             | red-brown ppt., insoluble in excess                | red-brown ppt., insoluble in excess                 |
| Cr <sup>3+</sup>             | green ppt., soluble in excess                      | grey-green ppt., insoluble in excess                |
| Zn <sup>2+</sup>             | white ppt., soluble in excess → colorless solution | white ppt., soluble in excess → colorless sol.      |
| Al <sup>3+</sup>             | white ppt., soluble in excess → colorless solution | white ppt., insoluble in excess                     |

| Test For gases                                                               |
|------------------------------------------------------------------------------|
| CO <sub>2</sub><br>turns limewater milky                                     |
| NH <sub>3</sub><br>turns damp red litmus paper blue                          |
| Cl <sub>2</sub><br>bleaches damp litmus paper                                |
| H <sub>2</sub><br>pops with a lighted splint                                 |
| O <sub>2</sub><br>relights a glowing splint                                  |
| SO <sub>2</sub><br>turns acidified aq. KMnO <sub>4</sub> purple to colorless |

| Anion                         | Test                                                             | Test result                                                                         |
|-------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| CO <sub>3</sub> <sup>2-</sup> | add dilute acid                                                  | effervescence, carbon dioxide produced                                              |
| Cl <sup>-</sup>               | acidify with dilute HNO <sub>3</sub> , add aq. AgNO <sub>3</sub> | white ppt.                                                                          |
| Br <sup>-</sup>               | acidify with dilute HNO <sub>3</sub> , add aq. AgNO <sub>3</sub> | cream ppt.                                                                          |
| I <sup>-</sup>                | acidify with dilute HNO <sub>3</sub> , add aq. AgNO <sub>3</sub> | yellow ppt.                                                                         |
| NO <sub>3</sub> <sup>-</sup>  | add aq. NaOH, then <u>aluminium</u> foil, warm                   | pungent NH <sub>3</sub> produced, turns damp red litmus paper blue                  |
| SO <sub>4</sub> <sup>2-</sup> | acidify, add aqueous barium nitrate                              | white ppt.                                                                          |
| SO <sub>3</sub> <sup>2-</sup> | add dil. HCl, warm, test for presence of SO <sub>2</sub>         | SO <sub>2</sub> produced turns acidified KMnO <sub>4</sub> from purple to colorless |

Flame test

Na<sup>+</sup> yellow  
Li<sup>+</sup> red  
K<sup>+</sup> lilac  
Cu<sup>2+</sup> blue-green  
Ca<sup>2+</sup> orange-red  
Ba<sup>2+</sup> light green

Rules

4 reactions:

1. Acid + metal → salt + hydrogen

metals below H do not react

do not use metals above Mg (explosive)

do not use Al (slow due to protective layer)

Type: displacement/redox

Observation: metal dissolves, bubbles of colorless gas

2. Acid + MO/MOH → salt + water

Type: neutralization

Observation: MO/MOH dissolves (if insoluble), solution becomes colored if transition

3. Acid + MCO<sub>3</sub> → salt + water + carbon dioxide

Type: neutralization

Observation: MCO<sub>3</sub> dissolves (if insoluble), bubbles of colorless gas, solution becomes colored if transition

4. Base + NH<sub>4</sub><sup>+</sup> salt → salt + water + ammonia

Type: displacement

Observation: base dissolves (if insoluble), pungent gas that turns damp red litmus paper blue

5 ways to differentiate strong vs. weak acid:

1. Measure pH using pH meter/universal indicator, strong acid will have lower pH
2. Measure electrical conductivity using voltmeter, strong acid will have higher conductivity
3. Add Zn or Mg, strong acid will produce faster bubbles
4. Add calcium carbonate (or any metal carbonate), strong acid will produce faster bubbles
5. Titrate with NaOH, strong acid will need a greater volume of NaOH for neutralization

3 ways to differentiate strong vs. weak alkali:

1. Measure pH using pH meter/universal indicator, strong alkali will have higher pH
2. Measure electrical conductivity using voltmeter, strong alkali will have higher conductivity
3. Titrate with HCl, strong alkali will need a greater volume of HCl for neutralization

Types of oxides

| Basic                                    | acidic                                                                                                 | amphoteric                                                                                                   | neutral                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------|
| Metal oxides                             | non-metal oxides                                                                                       | Zn, Al, Cr oxide                                                                                             | monoxides of non-metals  |
| MgO, CaO, Fe <sub>2</sub> O <sub>3</sub> | SO <sub>2</sub> , CO <sub>2</sub> , NO <sub>2</sub>                                                    | ZnO, Al <sub>2</sub> O <sub>3</sub> , Cr <sub>2</sub> O <sub>3</sub>                                         | CO, NO, H <sub>2</sub> O |
| Basic oxide + HCl ✓                      | acidic oxide + HCl ✗                                                                                   | amphoteric oxide + HCl ✓                                                                                     | neutral oxide + HCl ✗    |
| Basic oxide + NaOH ✗                     | acidic oxide + NaOH ✓                                                                                  | amphoteric oxide + NaOH ✓                                                                                    | neutral oxide + NaOH ✗   |
|                                          | e.g. CO <sub>2</sub> + limewater gives milky appearance of CaCO <sub>3</sub> (neutralization reaction) | e.g. how to separate CuO from mixture of CuO + ZnO? Add NaOH, only ZnO will dissolve, filter, CuO is residue |                          |



# Summary Chapter 9

Columns → group # → determines # of electrons in outermost shell = valency electrons  
 Rows → period # → determines # of shells

## Metals

Tendency to lose electrons  
 +ve ions

Metallic property ↑ down the group  
 Metallic property ↓ across the period

## non-metals

tendency to gain electrons  
 -ve ions (except H)

## Halogens - Group VII

### Properties:

1. Exists as diatomic molecules
2. Toxic
3. Color gets darker down the group
4. M.pt & b.pt ↑ down the group
5. Density ↑ down the group
6. Reactivity ↓ down the group
7. Reaction: displacement reaction, also redox

Rule

More reactive halogen displaces the less reactive halogen from its compound

### Examples:



Density increases down the group

M.pt & b.pt increase down the group

Reactivity decreases down the group

|                              |                     |
|------------------------------|---------------------|
| 9<br>F<br>fluorine<br>19     | Yellow<br>gas       |
| 17<br>Cl<br>chlorine<br>35.5 | yellow-green<br>gas |
| 35<br>Br<br>bromine<br>80    | red-brown<br>liquid |
| 53<br>I<br>iodine<br>127     | grey-black<br>solid |
| 85<br>At<br>astatine<br>-    | black<br>solid      |

## Alkali metals - Group I

### Physical properties:

1. Conduct electricity
2. Silvery grey in color
3. Malleable, ductile, Forms alloys
4. Density < 1 g/cm<sup>3</sup> (Floats)
5. Lower m.pt & b.pt than other metals
6. Soft, softness ↑ down the group

Reactivity increases down the group

M.pt & b.pt decrease down the group

Density increases down the group

Softness increases down the group

|                            |
|----------------------------|
| 3<br>Li<br>lithium<br>7    |
| 11<br>Na<br>sodium<br>23   |
| 19<br>K<br>potassium<br>39 |
| 37<br>Rb<br>rubidium<br>85 |
| 55<br>Cs<br>caesium<br>133 |

## Transition metals

### Physical Properties:

1. Harder
2. Higher melting point
3. Higher density (> 4g/cm<sup>3</sup>)

### Chemical Properties:

4. Forms colored compounds
5. Forms ions with variable oxidation states
6. Used as catalysts

## Noble gases, group VIII

### Properties:

1. Non-metals
2. Colorless gases
3. Stable due to Full outer electrons: 8 electrons
  - except He → 2 electrons
4. Unreactive, doesn't form compounds, monatomic
5. Occurs naturally in air in small amounts

M.pt & b.pt increase down the group

Density increases down the group

|                           |
|---------------------------|
| 2<br>He<br>helium<br>4    |
| 10<br>Ne<br>neon<br>20    |
| 18<br>Ar<br>argon<br>40   |
| 36<br>Kr<br>krypton<br>84 |
| 54<br>Xe<br>xenon<br>131  |

## Colors

Metals: silvery gray

Except: Cu & Au

Ionic compounds

Non-transition: white solid, colorless solution

KMnO<sub>4</sub> Purple

K<sub>2</sub>Cr<sub>2</sub>O<sub>7</sub> Orange

Carbon - black

3. 3 Reactions: all are redox reactions



Observation: Fizzing, solid disappears, Floats on water

Li: slow, Na: shoots across the beaker, K: sparks with flames, Rb: bursts into flames, Cs: explosive



Observation: metal tarnish quickly when exposed to air → must store under oil



# Properties of metals

| Physical properties                                               | Chemical properties                                  |
|-------------------------------------------------------------------|------------------------------------------------------|
| Conduct electricity & heat (no exception)                         | Forms +ve ions                                       |
| High melting & boiling point (transition: higher, group I: lower) | reacts with acid → salt + hydrogen                   |
| Hard & strong (except group I: soft)                              | reacts with oxygen → metal oxide (except Ag, Au, Pt) |
| Malleable & ductile                                               |                                                      |
| High density (transition > 4 gm/cm <sup>3</sup> , group I < 1)    |                                                      |
| Can Form alloys                                                   | Forms basic oxide                                    |
| Silvery gray in color (Cu: pink-brown, Au: yellow)                | (except Al, Zn, Cr: Forms amphoteric)                |

## Apparent unreactivity of aluminium

Due to the protective layer of Al<sub>2</sub>O<sub>3</sub> that acts as a barrier & protects Al From Further oxidation.  
Iron when exposed to air, Forms rust, which is Flaky & doesn't protect underlying iron

## Properties of alloys

- Harder (less malleable)
- Better appearance
- Less likely to corrode

## Uses of metals & alloys

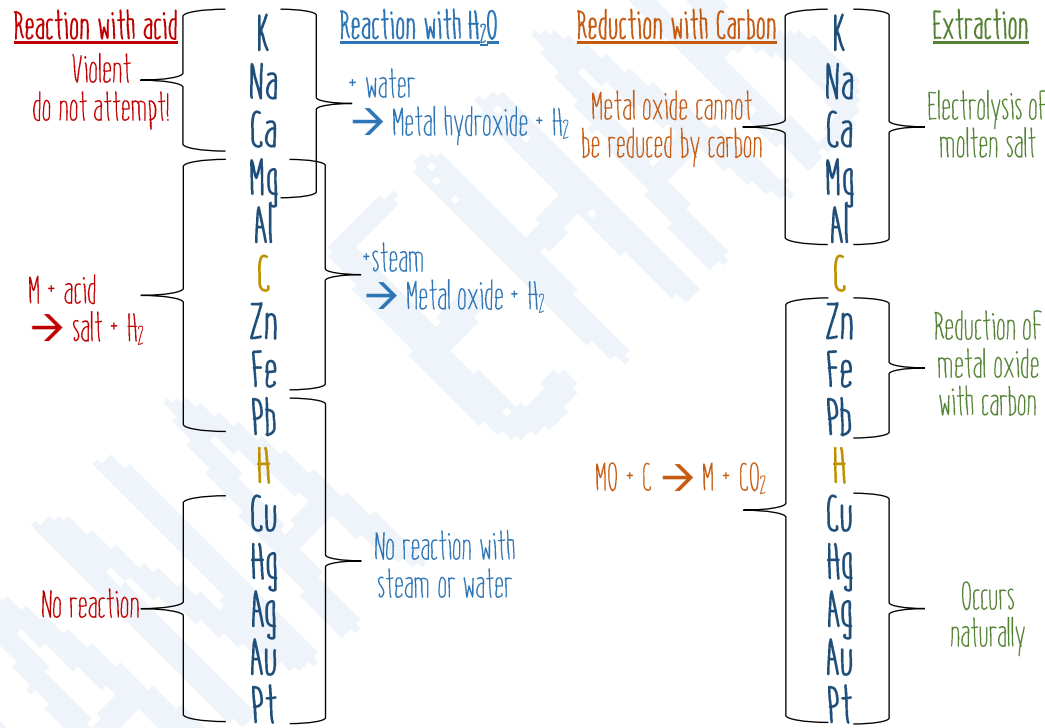
| Metal/alloy       | Use & reason                                                                                                                                                                                                                                                                                         |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Copper            | Electric wires: ductile good electric conductor, high melting point<br>Cooking utensils: good heat conductor, malleable<br>Making brass                                                                                                                                                              |
| Aluminium         | Aircraft: low density (also high strength, doesn't corrode)<br>Overhead cables: low density, doesn't corrode, conducts electricity (steel core)<br>Food containers & cans: doesn't corrode<br>Window Frames: doesn't corrode<br>Cooking utensils: doesn't corrode, conducts heat, high melting point |
| Zinc              | Galvanizing iron: more reactive than iron, thus protects iron                                                                                                                                                                                                                                        |
| Mild Steel        | Car bodies, machines, screws: hard                                                                                                                                                                                                                                                                   |
| High carbon steel | railway tracks, hammers, drills, cutting tools: harder than mild steel                                                                                                                                                                                                                               |
| Stainless steel   | cutlery, saucepans, surgery tools, pipes in chemical Factory: doesn't rust                                                                                                                                                                                                                           |

## Calcium compounds

| Compound                | Common name | Reaction of limewater with carbon dioxide                                                         | Type: Neutralization, precipitation |
|-------------------------|-------------|---------------------------------------------------------------------------------------------------|-------------------------------------|
| CaCO <sub>3</sub>       | Limestone   | CO <sub>2(g)</sub> + Ca(OH) <sub>2(aq)</sub> → CaCO <sub>3(s)</sub> + H <sub>2</sub> O(l)         |                                     |
| CaO                     | Lime        | Reaction of calcium oxide with water<br>CaO + H <sub>2</sub> O → Ca(OH) <sub>2</sub>              | Type: hydration, addition           |
| Ca(OH) <sub>2(s)</sub>  | Slaked lime | Uses of limestone: in extraction of Fe, making cement, making lime, neutralization of acidic soil |                                     |
| Ca(OH) <sub>2(aq)</sub> | Lime water  | Uses of lime & slaked lime: neutralization of acidic soil, desulfurization                        |                                     |

# Summary Chapter 10

## Reactivity series



## Thermal decomposition 4 Rules

- Metal carbonate → metal oxide + CO<sub>2</sub> except group 1
  - Metal hydroxide → metal oxide + H<sub>2</sub>O except group 1
  - Metal nitrate → metal oxide + NO<sub>2</sub> + O<sub>2</sub> except group 1
  - Metal nitrate (group 1) → metal nitrite + O<sub>2</sub>
- Obs: size ↓, color change if transition  
Obs: size ↓, color change if transition  
Obs: size ↓, color change if transition, brown fumes  
Obs: size ↓

## Extraction of iron

C + O<sub>2</sub> → CO<sub>2</sub> Exothermic: heats up the Furnace

CO<sub>2</sub> + C → 2CO

ZC + O<sub>2</sub> → 2CO

2Fe<sub>2</sub>O<sub>3</sub> + 3C → 4Fe + 3CO<sub>2</sub>

Fe<sub>2</sub>O<sub>3</sub> + 3CO → 2Fe + 3CO<sub>2</sub>

CaCO<sub>3</sub> → CaO + CO<sub>2</sub>

CaO + SiO<sub>2</sub> → CaSiO<sub>3</sub> Slag: Floats on molten iron & protects Fe from oxidation

4 raw materials:

- Ore: hematite (Fe<sub>2</sub>O<sub>3</sub>)
- Limestone
- Coke (carbon)
- Hot air

## Making steel

Cast iron From blast Furnace contains 4% carbon  
Cast iron poured into oxygen Furnace  
Oxygen is blown over iron to remove C, S, P, Si

C + O<sub>2</sub> → CO<sub>2(g)</sub> gas escapes  
S + O<sub>2</sub> → SO<sub>2(g)</sub> gas escapes  
Si/P + O<sub>2</sub> → oxides of Si/P (solid, removed by neutralization with CaO → Slag)

# Summary Chapter 11

## Chemical tests for water

- turns anhydrous  $\text{CuSO}_4$  From white to blue
- turns anhydrous  $\text{CoCl}_2$  From blue to pink

## Treatment of water

| Stage           | Reason                  |
|-----------------|-------------------------|
| 1. Filtration   | Remove insoluble solids |
| 2. Add carbon   | Remove tastes & odors   |
| 3. Chlorination | Kill bacteria           |

## Substances in water

1. dissolved oxygen: For aquatic life
2. metal cations: essential minerals, some are toxic
3. plastics: harms aquatic life
4. sewage: harmful microbes, cause disease
5. harmful microbes
6.  $\text{NO}_3^-$  &  $\text{PO}_4^{3-}$  From Fertilizers: deoxygenation of water & damage aquatic life

## Composition of air

- Nitrogen: 78%
- Oxygen: 21%
- Argon: 1%
- $\text{CO}_2$ : 0.04%
- Noble gases & water vapor

## Experiment to calculate % of $\text{O}_2$



Use cross multiplication to find x

|         |       |
|---------|-------|
| initial | final |
| 100     | 79    |
| x       | given |
| given   | x     |

## Separation of oxygen & nitrogen

Water vapor: removed by condensing into liquid

Acidic gases ( $\text{CO}_2$ ,  $\text{SO}_2$ ,  $\text{NO}_2$ ): removed by passing through NaOH

Fractional distillation of liquid air

## Uses of oxygen

1. Steel making
  2. Breathing apparatus
  3. Welding: oxygen mixed with  $\text{C}_2\text{H}_2$
- Advantage of  $\text{C}_2\text{H}_2$  (acetylene)
- Gas: easy to mix with  $\text{O}_2$
  - Produces hot flame

## Air pollution

| Pollutant          | Source                                                                  | Adverse effect                                      |
|--------------------|-------------------------------------------------------------------------|-----------------------------------------------------|
| CO                 | Incomplete combustion                                                   | Toxic                                               |
| $\text{SO}_2$      | Burning Fossil Fuels                                                    | Acid rain & respiratory problems                    |
| oxides of nitrogen | $\text{O}_2$ & $\text{N}_2$ in car engine react due to high temperature | Acid rain, respiratory problems, photochemical smog |
| $\text{CO}_2$      | Complete combustion                                                     | Greenhouse gas                                      |
| Particulates       | Incomplete combustion                                                   | Respiratory problems & cancer                       |
| $\text{CH}_4$      | Decomposition of vegetation & animal waste                              | Greenhouse gas                                      |

## Catalytic converters

Fitted in car exhausts to convert toxic gases into safe gases



## Acid rain

1. Damages trees
2. Damages limestone buildings
3. Acidify lakes
4. Increases acidity of soil
5. damage metal works

## How to reduce acid rain

1. Use catalytic converters
2. Using low-sulfur Fuels
3. Flue gas desulfurization with CaO

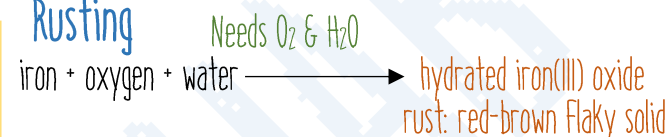
## Fertilizers

N P K  
Help crops grow

Farmer considerations:

1. Use carefully
2. Avoid rainy days
3. Use far from rivers

## Rusting



## Preventing rusting

### Cover with a barrier

- Paint: cars, railings, bridges
- Grease: machines
- Plastic
- Another metal: tin can
- Zinc: galvanization, protective even if scratched

### Electrochemical

Sacrificial protection E.g. Zn, Mg  
Zn is more reactive than Fe, thus, it loses electrons more readily than Fe and corrodes in preference to Fe  
Cathodic protection  
Connect to negative terminal

## Greenhouse gases $\text{CO}_2$ & $\text{CH}_4$

They absorb energy from earth & prevent energy from escaping

## Global warming

1. melting ice caps
2. extreme weather
3. Flooding
4. temperature increase
5. habitat loss

## How to reduce global warming

1. plant trees
2. reduce livestock farming
3. decrease use of fossil fuels
4. use  $\text{H}_2$  & renewable energy (solar, wind)

|            | $\text{CO}_2$                                                                                                                                                | $\text{CH}_4$                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Properties | Colorless, odorless gas<br>Insoluble in water<br>Denser than air<br>Doesn't burn, puts off fire                                                              | Colorless, odorless gas<br>Slightly soluble in water<br>Lighter than air<br>Burns in air |
| Uses       | Fire extinguisher<br>Fizzy drinks, dry ice                                                                                                                   | Fuel<br>Making organic chemicals                                                         |
| Sources    | Combustion, Respiration<br>acid + $\text{MCO}_3$ , Fermentation<br>$\text{MCO}_3$ decomposition<br>$\text{MO} + \text{C} \rightarrow \text{M} + \text{CO}_2$ | Decomposition of vegetation<br>Waste gases from animals<br>Natural gas is methane        |

# Petroleum Mixture of hydrocarbons Separated by Fractional distillation

| Mnemonic                        | Name                                                   | Use                                                                    |
|---------------------------------|--------------------------------------------------------|------------------------------------------------------------------------|
| Gas<br>1 - 4 carbons            | Ralph<br>ReFinery gas                                  | Bottled gas for cooking/heating                                        |
| Liquid<br>5 - 30 carbons        | Got<br>No<br>Ketchup<br>Naphtha<br>Kerosene (paraffin) | Petrol for cars<br>Making chemicals<br>Fuel for aircraft & home stoves |
| Thick Liquid<br>30 - 50 carbons | Dipping<br>For<br>Fuel oil                             | Fuel for diesel engines<br>Fuel for power stations, factories, ships   |
| Solid > 50 carbons              | Lunch<br>Break<br>Bitumen                              | Machine oil, waxes, polishes<br>Road surfaces & roofs                  |

Small molecules:

1. Few carbons
2. Low boiling points
3. Light in color
4. Flows easily
5. Flammable
6. Many uses

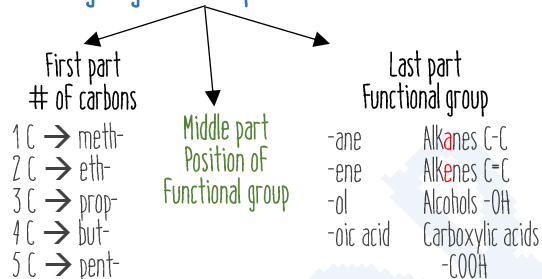
Large molecules:

1. Many carbons
2. High boiling points
3. Dark in color
4. Viscous
5. Doesn't burn easily
6. Few uses

## Disadvantages of using petroleum

1. Non-renewable resource
2. Production of greenhouse gas CO<sub>2</sub>
3. Acid rain, production of toxic gases

## Naming organic compounds

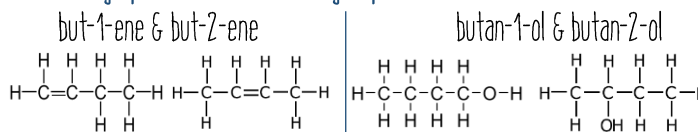


## Homologous series

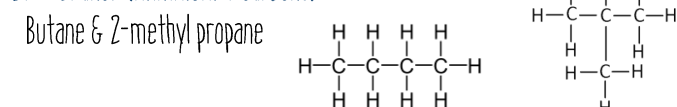
1. same general formula
2. same functional group
3. similar chemical properties
4. physical properties have a trend
5. consecutive members differ by -CH<sub>2</sub>-
6. can be prepared by similar methods

## Isomers

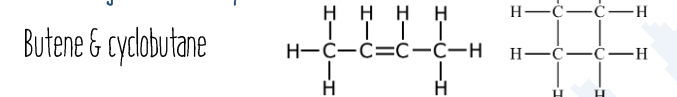
### 1. Change position of functional group



### 2. Branch (minimum 4 carbons)



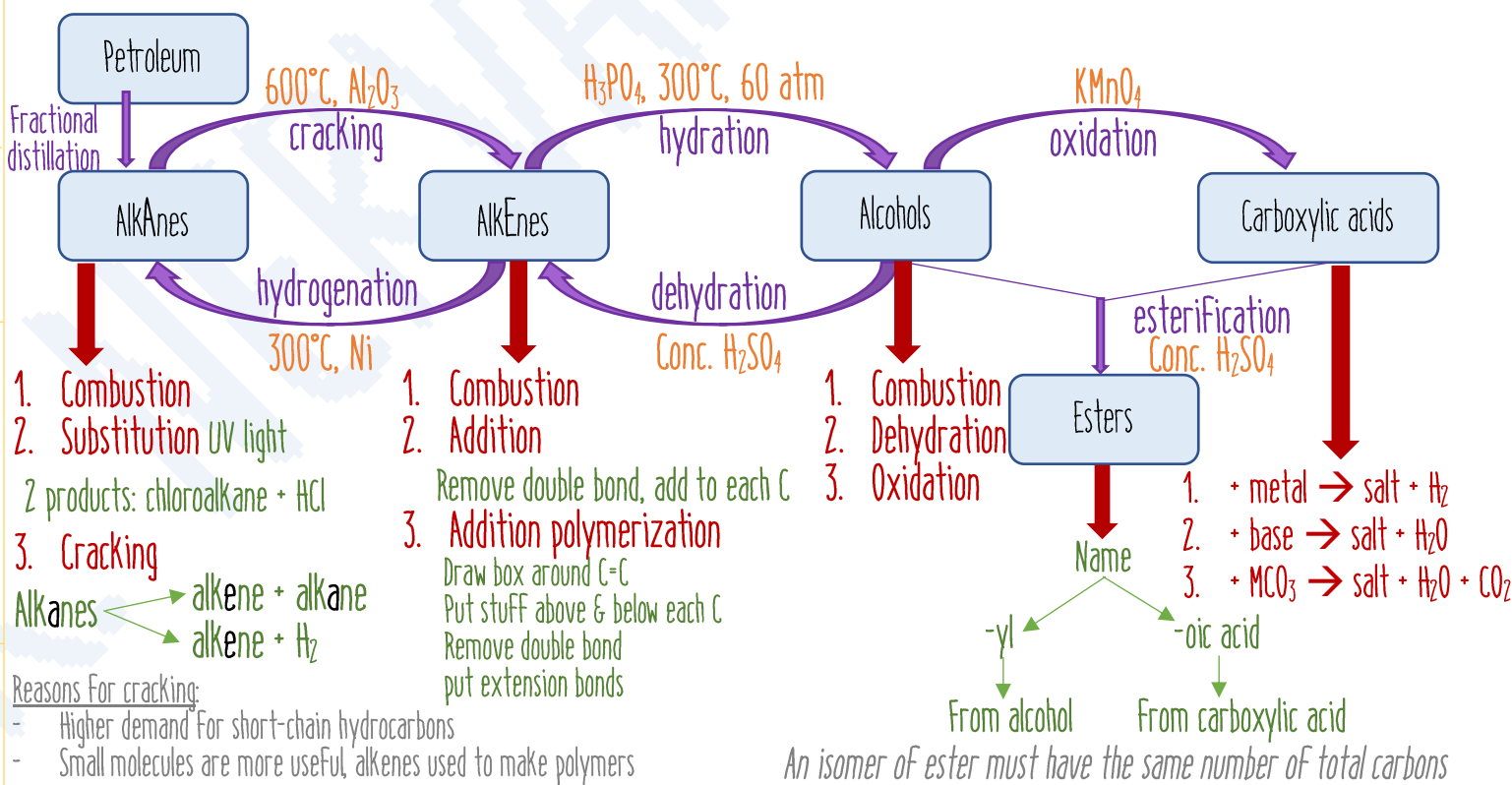
### 3. Change alkene to cycloalkane



## Problems of synthetic polymers (plastics)

Non-biodegradable: not decomposed by bacteria

1. Animals get trapped
2. Fills landfill sites
3. Produces poisonous vapors when burned
4. Litter and visual pollution
5. Sticks in throats of animals
6. Blocks drains & watercourses

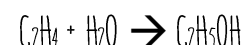


### Reasons For cracking:

- Higher demand for short-chain hydrocarbons
- Small molecules are more useful, alkenes used to make polymers

## Making ethanol

Catalytic addition of steam to ethene



H<sub>3</sub>PO<sub>4</sub>, 300°C, 60 atm

1. Fast
2. Pure product
3. Continuous

1. Uses non-renewable resource
2. Costly (high temp & pressure)

Fermentation



37°C, yeast (enzyme), no oxygen, aqueous glucose

1. Uses renewable resource
2. cheap

1. Slow
2. Low yield
3. Impure products

## Condensation polymerization

| Nylon                                   | PET                                  | Protein                       |
|-----------------------------------------|--------------------------------------|-------------------------------|
| Polyamide linkage                       | Polyester linkage                    | Polyamide/polypeptide linkage |
| Monomers: di-carboxylic acid + di-amine | Monomers: di-carboxylic acid + di-ol | Monomers: Amino acids         |
|                                         |                                      |                               |